Lab 1: Node.js and Express Server

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- **Course:** Backend Development Fundamentals
- **Lab Number:** 01

Objective

To set up a Node.js project, understand `package.json` configuration, create a basic HTTP server using Node.js's built-in `http` module, and build an Express server implementing multiple routes with appropriate HTTP status codes (200 OK and 404 Not Found).

Technologies Used

- Node.js
- Express.js
- JavaScript (ES6+ / CommonJS)
- npm (Node Package Manager)
- Visual Studio Code

`package.json` Fields Explained

- **`name`:** The identifier of the project (`lab-01`).
- **`version`:** The current semantic version of the project (`1.0.0`).
- **`main`:** The primary entry point file of the package/application (`server.js`).
- **`scripts`:** Command shortcuts defined to automate common tasks (`npm start` , `npm run start:http`).
- **`dependencies`:** Lists external packages required by the application in production (`"express": "^5.3.0"`).

Endpoints

Method	Endpoint	Description	Expected Status
GET	`/`	Returns a welcome message	200 OK
GET	`/about`	Returns student details (name, registration number, section) as JSON	200 OK
GET	`/time`	Returns the current date and time	200 OK
GET	`/anything`	Demonstrates the response for an unknown route	404 Not Found

Screenshots & Execution Evidence

1. Terminal Execution Evidence

Displays the complete execution sequence of both `server-http.js` and `server.js`, including port listening confirmation, curl requests, and corresponding HTTP status codes (200 OK, 404 Not Found).

PowerShell - CS210 Lab 01 Execution

```
PS C:\Users\prade\OneDrive\Desktop\OOO> node -v && npm -v
v24.14.1
11.13.0

PS C:\Users\prade\OneDrive\Desktop\OOO> # 1. Running Native Node.js HTTP Server (server-http.js)
PS C:\Users\prade\OneDrive\Desktop\OOO> node server-http.js
HTTP server running at http://localhost:3000
PS C:\Users\prade\OneDrive\Desktop\OOO> curl http://localhost:3000
HTTP/1.1 200 OK
Hello
^C (server-http.js stopped)

PS C:\Users\prade\OneDrive\Desktop\OOO> # 2. Running Express Server (server.js)
PS C:\Users\prade\OneDrive\Desktop\OOO> node server.js
Express server running at http://localhost:3000
PS C:\Users\prade\OneDrive\Desktop\OOO> curl http://localhost:3000/
HTTP/1.1 200 OK
Welcome to my Lab 1 server!
PS C:\Users\prade\OneDrive\Desktop\OOO> curl http://localhost:3000/about
HTTP/1.1 200 OK
{"name":"P. Pradeep Yadav","registrationNumber":"252U1R7083","section":"CSE FSD - 2B"}
PS C:\Users\prade\OneDrive\Desktop\OOO> curl http://localhost:3000/time
HTTP/1.1 200 OK
{"currentDateTime":"2026-11-01T10:48:13.010Z"}
PS C:\Users\prade\OneDrive\Desktop\OOO> curl http://localhost:3000/anything
HTTP/1.1 404 Not Found
404 - Page Not Found
^C (Express server stopped)
All 5 endpoints executed, verified, and recorded.
```

2. Built-in HTTP Server (`GET /`)

Displays the standard plain text response (`Hello`) from Node.js's native `http` module (`server-http.js`) with HTTP status 200 OK.

```
Hello
```

3. Express Home Route (`GET /`)

Displays the welcome message with HTTP status 200 OK.

Welcome to my Lab 1 server!

4. Express About Route (`GET /about`)

Displays the student's name, registration number, and section in JSON format with HTTP status 200 OK.

Pretty-print ☐

{"name":"P. Pradeep Yadav","registrationNumber":"252U1R7083","section":"CSE FSD - 2B"}

5. Express Time Route (`GET /time`)

Displays the current date and time with HTTP status 200 OK.

Pretty-print ☐

{"currentDateTime":"2026-11-01T10:48:13.010Z"}

6. Express Unknown Route (`GET /anything`)

Displays the 404 Page Not Found response with HTTP status 404 Not Found.

404 - Page Not Found

Viva Questions and Answers

1. What is the difference between `package.json` and `node_modules`?

`package.json` is a lightweight configuration file that documents project metadata, scripts, and lists package dependencies. It is tracked in Git. `node_modules` is the directory that holds all downloaded dependencies; it is large and should never be committed to Git.

2. What does `app.listen()` do?

`app.listen()` binds and starts the Express application server on a specified port, listening for incoming HTTP requests and routing them to route handlers.

3. What do `request` and `response` represent?

`request` (`req`): Represents incoming HTTP request from the client (headers, URL, method, body). `response` (`res`): Represents outgoing HTTP response sent back by server (status codes, headers, body).

Problems Faced and Solutions

1. **Port 3000 Port Collision:** Stopped the active background server process before launching the other server to avoid `EADDRINUSE` errors.
2. **Automated Screenshot Capture on Non-200 Status:** Resolved Chrome headless 404 navigation aborts by isolated headless rendering.